Shapley Value and Energy-Aware Dual Scheduling for Federated Learning with Secure Gradient Transmission

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***Abstract*—Federated learning (FL) enables collaborative model training across distributed edge devices without sharing raw data. However, existing client selection mechanisms face challenges in heterogeneous environments: non-IID data distribution, limited energy resources, and insufficient quantification of individual client contributions. While recent works explore gradient-based and channel-aware scheduling for over-the-air computation, they focus on instantaneous metrics and short-term optimization, overlooking long-term sustainability and accurate marginal contribution assessment.**

This paper proposes a Shapley value and energy-aware dual scheduling mechanism for federated learning with secure gradient transmission. We employ Shapley values to quantify each client's marginal contribution using an MC-Shapley approximation (Monte Carlo permutation sampling) with exponential smoothing for dynamic adaptation in non-IID scenarios. Energy consumption is modeled through wireless channel conditions ($E = \sigma^2/|h|^2$) with cumulative residual energy tracking. A Lyapunov optimization framework dynamically adjusts weights between Shapley values and energy scores. Gradient updates are protected during transmission using AES-256-GCM authenticated encryption, ensuring confidentiality and integrity under an honest-but-curious server threat model without degrading model accuracy.

Extensive simulations on CIFAR-10 with 100 clients demonstrate that our mechanism achieves higher test accuracy and faster convergence compared to random selection and Power of Choice baselines under highly non-IID data distribution (Dirichlet $\alpha = 0.1$), while maintaining energy efficiency and secure gradient transmission.

***Index Terms*—Federated learning, client selection, Shapley value, energy-aware scheduling, gradient encryption, AES-256-GCM, non-IID data, Lyapunov optimization**

I. Introduction

Federated learning enables collaborative model training across distributed edge devices without sharing raw data. However, existing client selection mechanisms face challenges in heterogeneous environments: non-IID data distribution, limited energy resources, and insufficient quantification of individual client contributions.

This paper proposes a Shapley value and energy-aware dual scheduling mechanism with secure gradient transmission. Our main contributions are:

- Shapley value-based contribution assessment using MC-Shapley approximation (Monte Carlo permutation sampling) with exponential smoothing for dynamic adaptation in non-IID scenarios
- Energy sustainability modeling through wireless channel conditions with cumulative residual energy tracking
- Lyapunov optimization framework for dynamic weight adjustment between accuracy and energy objectives
- AES-256-GCM authenticated encryption for gradient transmission, providing confidentiality and integrity under an honest-but-curious server threat model without accuracy degradation
- Comprehensive evaluation on CIFAR-10 with 100 clients under extreme non-IID conditions (Dirichlet $\alpha = 0.1$)

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II. Related Work

A. Federated Learning and Client Selection

FedAvg and its variants have been widely studied. Random selection ignores client heterogeneity, while biased selection strategies aim to improve convergence.

B. Contribution-Based Methods

Gradient-based methods use gradient norms as importance indicators. Power of Choice selects clients based on local loss values. Shapley value methods provide fair contribution assessment but lack integration with resource constraints.

C. Resource-Aware Federated Learning

Energy-constrained federated learning considers battery-powered devices. Channel-aware scheduling optimizes communication efficiency. Our work combines contribution assessment with long-term energy sustainability.

D. Privacy-Preserving Federated Learning

Privacy-preserving mechanisms in federated learning range from differential privacy (DP), which adds calibrated noise to gradients, to cryptographic approaches such as secure multi-party computation (SMPC) and homomorphic encryption (HE). While DP provides formal information-theoretic guarantees, it inevitably degrades model accuracy. Cryptographic methods preserve exact gradient values but incur computational overhead. We adopt AES-256-GCM authenticated encryption at the transmission layer, which provides confidentiality and integrity against honest-but-curious servers and external eavesdroppers without any accuracy loss, offering a practical trade-off between security and utility.

TABLE I
LIST OF MAIN NOTATIONS

Parameter	Definition
	Federated Learning
N	Total number of clients
K	Number of selected clients per round
T	Total number of communication rounds
E	Number of local training epochs
$\mathcal{N}$	Set of all clients $\{1, 2, \ldots, N\}$
$\mathcal{S}_t$	Subset of selected clients in round t
$\mathcal{D}_n$	Local dataset of client n
p_n	Data proportion of client n
w_t	Global model parameter in round t
Δw_n	Model update of client n
$F(w)$	Global loss function
$F_n(w)$	Local loss function of client n
η	Learning rate
α	Dirichlet concentration parameter / Exponential smoothing parameter
	Shapley Value
$\phi_n(t)$	Shapley value of client n in round t
$v(\mathcal{S})$	Utility function of coalition $\mathcal{S}$
$\bar{\phi}_n(t)$	Smoothed Shapley value
	Energy and Lyapunov
$E_n(t)$	Energy consumption of client n in round t
$E_{\text{remain},n}(t)$	Remaining energy of client n
$h_n(t)$	Channel gain between client n and server
σ^2	Noise power
$Q_n(t)$	Lyapunov virtual queue of client n
$\hat{Q}_n(t)$	Normalized virtual queue
V	Lyapunov control parameter
w_ϕ	Weight for Shapley value
w_E	Weight for energy score
	Security
k_n	AES-256 session key of client n
r_n	Fresh 96-bit nonce per upload
τ_n	128-bit GCM authentication tag

III. SYSTEM MODEL

A. Federated Learning Framework

We consider a federated learning system consisting of one central parameter server and N edge devices (clients), denoted by the set $\mathcal{N} = \{1, 2, \ldots, N\}$. Each client n possesses a local private dataset $\mathcal{D}_n$ with $|\mathcal{D}_n|$ samples, and the goal is to collaboratively train a global model $w \in \mathbb{R}^d$ that minimizes the global loss function:

$$F(w) = \sum_{n=1}^{N} p_n F_n(w) \tag{1}$$

where $p_n = |\mathcal{D}_n| / \sum_{i=1}^{N} |\mathcal{D}_i|$ represents the data proportion of client n (with $\sum_{n=1}^{N} p_n = 1$), and $F_n(w) = \mathbb{E}_{(x,y)\sim\mathcal{D}_n}[\ell(w; x, y)]$ is the local loss function with $\ell(\cdot)$ being the loss function (e.g., cross-entropy for classification tasks).

Training Protocol: The federated learning process proceeds in T communication rounds. In each round t:

1) **Global Model Broadcasting:** The server broadcasts the current global model w_t to a selected subset of clients $\mathcal{S}_t \subseteq \mathcal{N}$ with $|\mathcal{S}_t| = K \ll N$.
2) **Local Training:** Each selected client $n \in \mathcal{S}_t$ performs local training for E epochs using stochastic gradient descent (SGD) on its local dataset $\mathcal{D}_n$, starting from w_t. The local model update is:

$$w_n^{t+1} = w_t - \eta \nabla F_n(w_t) \tag{2}$$

where η is the learning rate. The model difference is $\Delta w_n = w_n^{t+1} - w_t$.
3) **Model Upload:** Each client $n \in \mathcal{S}_t$ uploads its model update Δw_n to the server.
4) **Global Aggregation:** The server aggregates the received updates using weighted FedAvg, where each client's contribution is proportional to its local dataset size:

$$w_{t+1} = \sum_{n\in\mathcal{S}_t} \frac{|\mathcal{D}_n|}{\sum_{i\in\mathcal{S}_t} |\mathcal{D}_i|} w_n^{t+1} \tag{3}$$

The key challenge lies in the client selection strategy for determining $\mathcal{S}_t$ at each round, which significantly impacts convergence speed, model accuracy, and resource efficiency.

B. Non-IID Data Distribution

In real-world federated learning scenarios, data across clients is typically non-independent and identically distributed (non-IID), meaning that the local data distributions $\mathcal{D}_n$ differ significantly across clients. This heterogeneity poses substantial challenges to model convergence and performance.

Dirichlet Distribution Modeling: We model the non-IID data distribution using the Dirichlet distribution, a widely adopted approach in federated learning research. Specifically, for a classification task with C classes, the class distribution of client n is sampled as:

$$p_n = (p_{n,1}, p_{n,2}, \ldots, p_{n,C}) \sim \text{Dir}(\alpha) \tag{4}$$

where $\alpha > 0$ is the concentration parameter controlling the degree of heterogeneity:

- $\alpha \to \infty$: The distribution approaches uniform, resulting in i.i.d. data across clients.
- $\alpha \to 0$: The distribution becomes highly skewed, with each client possessing data from only a few classes (extreme non-IID).
- $\alpha = 0.1$: Our experimental setting, representing extreme heterogeneity where clients have highly imbalanced class distributions.

Impact on Convergence: Non-IID data introduces gradient divergence across clients, leading to slower convergence compared to i.i.d. settings, increased variance in gradient estimates, and potential model drift where the global model favors clients with dominant data distributions. This motivates the need for intelligent client selection mechanisms that account for data heterogeneity and prioritize clients with high marginal contributions to the global model.

C. Wireless Channel and Energy Model

Edge devices in federated learning systems are typically battery-powered and communicate with the server over wireless channels. We model both the wireless channel conditions and energy consumption to capture the resource constraints in practical deployments.

Channel Model: We adopt the Rayleigh fading channel model, commonly used for non-line-of-sight wireless communications. The channel coefficient between client n and the server at round t is modeled as:

$$h_n(t) \sim \mathcal{CN}(0, 1) \tag{5}$$

where $\mathcal{CN}(0,1)$ denotes the complex Gaussian distribution with zero mean and unit variance. The channel power gain is $|h_n(t)|^2$, which follows an exponential distribution.

Energy Consumption Model: The total energy consumed by client n in round t comprises transmission energy and computation energy:

$$E_n(t) = E_{\text{trans},n}(t) + E_{\text{comp},n}(t) \tag{6}$$

The transmission energy is inversely proportional to the channel quality:

$$E_{\text{trans},n}(t) = \min\left(\frac{\sigma^2}{|h_n(t)|^2}, E_{\max}\right) \tag{7}$$

where σ^2 represents the noise power and $E_{\max}$ is an upper bound to prevent extreme values from poor channel conditions.

The computation energy for local training is:

$$E_{\text{comp},n}(t) = \kappa \cdot f^2 \cdot C_{\text{cyc}} \cdot D_n \tag{8}$$

where κ is the effective switched capacitance coefficient of the CPU, f is the CPU frequency, C_{cyc} is the number of CPU cycles per training sample, and $D_n = |\mathcal{D}_n|$ is the local dataset size.

Residual Energy Tracking: Each client n maintains a cumulative residual energy $E_{\text{remain},n}(t)$, initialized to E_{init} at the beginning of training. When client n is selected in round t (denoted by indicator $a_n(t) = 1$), its residual energy is updated as:

$$E_{\text{remain},n}(t+1) = E_{\text{remain},n}(t) - a_n(t) \cdot E_n(t) \tag{9}$$

For unselected clients ($a_n(t) = 0$), the residual energy remains unchanged.

Energy Constraint: To ensure long-term sustainability, we impose an energy threshold constraint:

$$E_{\text{remain},n}(t) \geq E_{\text{threshold}} \tag{10}$$

Clients whose residual energy falls below $E_{\text{threshold}}$ are excluded from the candidate pool for selection.

D. Security Model

While federated learning avoids direct data sharing, model updates can still leak sensitive information about clients' private data through gradient inversion attacks. To protect gradient confidentiality and integrity during transmission, we integrate AES-256-GCM authenticated encryption into the federated learning pipeline.

Threat Model: We adopt the honest-but-curious (semi-honest) server model, which is the standard assumption in federated learning security literature. The server faithfully executes the protocol but may attempt to infer private information from received model updates. External eavesdroppers are also considered as adversaries.

AES-256-GCM Encryption: Each client n holds an independent 256-bit session key k_n established during initialization. Before uploading its model update Δw_n, client n encrypts it using AES-GCM:

$$c_n = \text{AES-GCM}_{k_n}(\text{Serialize}(\Delta w_n), \text{nonce}_n) \tag{11}$$

where nonce_n is a fresh 96-bit random nonce generated for each upload, and the output $c_n = (\text{ciphertext}, \text{tag})$ includes a 128-bit authentication tag.

Server-Side Processing: Upon receiving c_n, the server decrypts and authenticates:

$$\Delta w_n = \text{AES-GCM-Decrypt}_{k_n}(c_n, \text{nonce}_n) \tag{12}$$

The server uses Δw_n ephemerally for Shapley value computation and model aggregation, then immediately destroys the plaintext. Only the ciphertext c_n is retained for archival purposes.

Security Properties: AES-256-GCM provides: (1) *confidentiality*: ciphertext is computationally indistinguishable from random under IND-CPA, protecting gradient contents from eavesdroppers; (2) *integrity*: the authentication tag detects any tampering with the ciphertext; (3) *zero accuracy loss*: encryption is applied post-training and decrypted before aggregation, leaving model training completely unaffected.

Comparison with Differential Privacy: Unlike CDP which adds Gaussian noise to gradients and inevitably degrades model accuracy, AES-256-GCM operates purely at the transmission layer. The privacy guarantee is cryptographic rather than information-theoretic, and is incompatible with honest-but-curious server computation on ciphertext. This trade-off is appropriate when the server is semi-trusted and the primary concern is protecting gradient transmission from external observers or post-hoc data leakage.

IV. Problem Formulation

A. Baseline Problem: Convergence Optimization

The fundamental objective of federated learning is to minimize the global loss function through iterative client selection and model aggregation. We first formulate the baseline optimization problem:

Problem P1 (Convergence-Only):

$$\min_{\{a_n(t)\}} \quad F(w_T) \tag{13}$$

$$\text{s.t.} \quad \sum_{n=1}^{N} a_n(t) = K, \quad \forall t \in \{1, \ldots, T\}, \tag{14a}$$

$$a_n(t) \in \{0, 1\}, \quad \forall n, t, \tag{14b}$$

where $a_n(t)$ is the binary selection indicator for client n at round t, and w_T is the global model after T rounds.

Limitation: Problem P1 ignores client heterogeneity and resource constraints. Random selection (FedAvg) treats all clients equally, leading to suboptimal convergence in non-IID settings and potential energy depletion for frequently selected clients.

B. Contribution-Aware Problem with Energy Constraints

To address client heterogeneity, we introduce Shapley values to quantify marginal contributions and add energy sustainability constraints:

Problem P2 (Shapley + Energy):

$$\max_{\{a_n(t)\}} \quad \sum_{t=1}^{T}\sum_{n=1}^{N} a_n(t)\cdot\varphi_n(t) \tag{15a}$$

$$\text{s.t.} \quad \sum_{n=1}^{N} a_n(t) = K, \quad \forall t, \tag{15b}$$

$$\frac{1}{T}\sum_{t=1}^{T} a_n(t)E_n(t) \le \bar{E}_n, \quad \forall n, \tag{15c}$$

$$a_n(t) \in \{0,1\}, \tag{15d}$$

where $\varphi_n(t)$ is the Shapley value of client n at round t, $E_n(t) = \sigma^2/|h_n(t)|^2$ is the transmission energy consumption, and $\bar{E}_n$ is the average energy budget.

Shapley Value Definition: For client n, the Shapley value quantifies its marginal contribution:

$$\varphi_n(t) = \sum_{S\subseteq\mathcal{N}\setminus\{n\}} \frac{|S|!(N-|S|-1)!}{N!}[v(S\cup\{n\}) - v(S)] \tag{16}$$

where $v(S)$ is the utility function defined as the negative validation loss of the aggregated model from coalition S.

Limitation: Problem P2 couples decisions across all rounds through the average energy constraint, making it computationally intractable for online decision-making.

C. Lyapunov-Based Online Optimization

To enable online decision-making, we reformulate P2 using Lyapunov optimization, which decouples the long-term energy constraint into per-round decisions.

Virtual Queue Construction: For each client n, define a virtual energy queue that tracks cumulative energy constraint violations:

$$Q_n(t+1) = \max\{Q_n(t) + E_n(t) - \bar{E},\ 0\} \tag{17}$$

where $E_n(t)$ is the actual energy consumed by client n in round t (zero if not selected), and $\bar{E}$ is the per-round energy budget. The queue increases when a client's energy consumption exceeds the budget, signaling that the client has been consuming excessive energy.

Lyapunov Function:

$$L(Q(t)) = \frac{1}{2}\sum_{n=1}^{N} Q_n^2(t) \tag{18}$$

Problem P3 (Per-Round Optimization): At each round t, minimize the drift-plus-penalty:

$$\min_{\{a_n(t)\}} \quad \Delta(Q(t)) + V\cdot\mathbb{E}[-\text{Utility}(t)] \tag{19a}$$

$$\text{s.t.} \quad \sum_{n=1}^{N} a_n(t) = K, \tag{19b}$$

$$a_n(t) \in \{0,1\}, \tag{19c}$$

where $\Delta(Q(t)) = \mathbb{E}[L(Q(t+1)) - L(Q(t))]$ is the Lyapunov drift, and $V > 0$ controls the tradeoff between queue stability (energy) and utility (accuracy).

D. Practical Solution: Composite Scoring

To solve P3 efficiently, we design a composite scoring function derived from the drift-plus-penalty framework. By upper-bounding the Lyapunov drift term, the per-round optimization reduces to maximizing the following score for each client:

Composite Score:

$$\text{Score}_n(t) = V\cdot\hat{\varphi}_n(t) - \hat{Q}_n(t)\cdot\hat{E}_n(t) \tag{20}$$

where $\hat{\varphi}_n(t) = (\varphi_n(t) - \varphi_{\min}(t))/(\varphi_{\max}(t) - \varphi_{\min}(t))$ is the min-max normalized Shapley value, $\hat{E}_n(t) = E_n(t)/E_{\max}(t)$ is the normalized energy consumption, and $\hat{Q}_n(t) = Q_n(t)/\max_i Q_i(t)$ is the normalized per-client virtual queue. Normalizing the queue prevents its magnitude from overwhelming the Shapley term as energy violations accumulate over time. The parameter $V > 0$ controls the tradeoff: larger V prioritizes Shapley-based client quality, while the normalized queue $\hat{Q}_n(t)$ automatically penalizes clients that have accumulated excessive energy consumption.

Greedy Selection: At each round, select the top-K clients with highest composite scores among energy-available clients.

Secure Transmission: Encrypt gradient updates using AES-256-GCM before uploading to the server; the server decrypts ephemerally for Shapley computation and aggregation, then immediately destroys plaintext.

V. Dual Scheduling Mechanism

A. Shapley Value Computation

Computational Challenge: Exact Shapley value computation requires evaluating all 2^N possible coalitions, which is computationally prohibitive. We address this using MC-Shapley, a Monte Carlo permutation sampling approach combined with exponential smoothing.

MC-Shapley Approximation: The Shapley value for client n in round t is defined over all possible coalitions:

$$\varphi_n^{\text{current}}(t) = \sum_{S\subseteq\mathcal{S}_t\setminus\{n\}} \frac{|S|!(|\mathcal{S}_t| - |S| - 1)!}{|\mathcal{S}_t|!}[v(S\cup\{n\}) - v(S)] \tag{21}$$

where $v(S)$ denotes the utility (negative validation loss) of the aggregated model from coalition S. This exact formulation requires enumerating all $2^{|\mathcal{S}_t|}$ coalitions, which is computationally prohibitive. To address this, our MC-Shapley method adopts *Monte Carlo permutation sampling* as a practical approximation: in each iteration m, we randomly sample a permutation π_m of the selected clients, then compute the marginal contribution of each client $n = \pi_m(j)$ as:

$$\Delta_n^{(m)} = v(\{\pi_m(1),\ldots,\pi_m(j)\}) - v(\{\pi_m(1),\ldots,\pi_m(j-1)\}) \tag{22}$$

The Shapley value is estimated as the average marginal contribution over M random permutations:

$$\varphi_n^{\text{current}}(t) \approx \frac{1}{M}\sum_{m=1}^{M}\Delta_n^{(m)} \tag{23}$$

This reduces the computational complexity from $O(2^K)$ to $O(K\times M)$ where $M \ll 2^K$. We further apply two truncation

strategies to accelerate computation: (1) *inter-round truncation*: if $|v(w_t) - v(w_{t-1})| < \epsilon$, skip Shapley computation; (2) *intra-round truncation*: within a permutation, stop evaluating when $|v(S_j) - v(\mathcal{S}_t)| < \epsilon$.

Critical Innovation — Exponential Smoothing: A key design choice is the use of exponential smoothing to update Shapley values across rounds, balancing responsiveness to recent contributions with stability from historical observations:

$$\varphi_n(t) = \alpha \cdot \varphi_n(t-1) + (1-\alpha) \cdot \varphi_n^{\text{current}}(t) \tag{24}$$

where $\alpha \in [0, 1]$ is the smoothing parameter. A larger α gives more weight to historical values (noise filtering), while a smaller α prioritizes recent observations (fast adaptation). We set $\alpha = 0.5$ to balance noise filtering and rapid adaptation to changing client contributions in non-IID settings.

B. Energy-Aware Scoring

For each client n, we compute a normalized energy score:

$$s_{\text{energy},n}(t) = \frac{E_{\text{remain},n}(t)}{E_{\text{init}}} \tag{25}$$

Before scoring, we filter out clients whose energy has fallen below the threshold:

$$\mathcal{A}(t) = \{n \mid E_{\text{remain},n}(t) \geq E_{\text{threshold}}\} \tag{26}$$

At each round, we generate channel coefficients $h_n(t) \sim \mathcal{CN}(0, 1)$ for all clients and compute energy consumption as defined in Section III.

C. Dual Scheduling Mechanism

For each eligible client $n \in \mathcal{A}(t)$, we compute the composite score derived from Lyapunov drift-plus-penalty optimization:

$$\text{Score}_n(t) = V \cdot \hat{\varphi}_n(t) - \hat{Q}_n(t) \cdot \hat{E}_n(t) \tag{27}$$

where $\hat{\varphi}_n(t)$ is the min-max normalized Shapley value, $\hat{E}_n(t)$ is the normalized energy consumption, and $\hat{Q}_n(t) = Q_n(t)/\max_i Q_i(t)$ is the normalized per-client virtual queue. Normalizing $Q_n(t)$ prevents the queue magnitude from overwhelming the Shapley value term as training progresses.

Top-K Selection: We employ a greedy selection strategy: (1) Sort all eligible clients by $\text{Score}_n(t)$ in descending order; (2) Select the top-K clients: $\mathcal{S}_t = \text{top-}K(\mathcal{A}(t))$.

D. Lyapunov-Based Dynamic Weight Adjustment

To balance Shapley-based client quality and energy sustainability, we adopt Lyapunov optimization with per-client virtual queues. For each client n, we maintain a virtual queue $Q_n(t)$ that tracks cumulative energy budget violations.

After each round t, for selected clients, we update:

$$Q_n(t+1) = \max\left\{Q_n(t) + E_n(t) - \bar{E},\ 0\right\} \tag{28}$$

where $E_n(t)$ is the actual energy consumed by client n in round t, and $\bar{E}$ is the per-round energy budget. When a client's energy consumption consistently exceeds the budget, its queue value $Q_n(t)$ grows, increasing the energy penalty in the scoring function and naturally steering selection away from energy-intensive clients. This per-client penalty mechanism adapts individually to each client's energy consumption pattern, providing finer-grained control than global weight adjustment.

E. AES-256-GCM Gradient Encryption

To protect gradient updates during transmission under the honest-but-curious server threat model, we integrate AES-256-GCM authenticated encryption as a transmission-layer security mechanism.

Per-Client Key Management: At system initialization, each client n is assigned an independent 256-bit AES session key k_n. Keys are established once and reused across all rounds:

$$k_n \xleftarrow{\$} \{0,1\}^{256}, \quad \forall n \in \mathcal{N} \tag{29}$$

Encrypt-then-Upload: After local training, client n serializes its model update and encrypts it with a fresh nonce per round:

$$(c_n, \tau_n) = \text{AES-GCM}_{k_n}(\text{Serialize}(\Delta w_n),\ r_n) \tag{30}$$

where $r_n \xleftarrow{\$} \{0,1\}^{96}$ is a fresh 96-bit nonce, c_n is the ciphertext, and $\tau_n \in \{0,1\}^{128}$ is the GCM authentication tag.

Server-Side Ephemeral Decryption: The server decrypts, verifies the tag, uses the plaintext for Shapley computation and FedAvg aggregation, then immediately destroys the plaintext reference:

$$\Delta w_n = \text{AES-GCM-Dec}_{k_n}(c_n, \tau_n, r_n), \quad \text{then } \texttt{del}(\Delta w_n) \tag{31}$$

Only ciphertext (c_n, τ_n, r_n) is retained for audit purposes. This protocol ensures that no plaintext gradient persists on the server beyond the duration of a single round's computation.

F. Complete Algorithm Framework

VI. Theoretical Analysis

A. Computational Complexity

1) Per-Round Complexity: The computational complexity of our Dual Scheduling mechanism per round is:

Shapley Value Computation:

- MC-Shapley with M Monte Carlo iterations: $O(K \times M)$
- For $K = 10$ selected clients and $M = 5$ iterations: $O(50)$
- Significantly lower than exact Shapley: $O(2^K) = O(1024)$

Energy and Channel Simulation:

- Channel gain generation: $O(N)$
- Energy consumption calculation: $O(K)$
- Virtual queue update: $O(N)$

Client Selection:

- Score computation: $O(N)$
- Top-K selection: $O(N \log K)$

Total per-round complexity: $O(N \log K + K \times M)$, which is practical for large-scale federated learning.

Algorithm 1 Dual Scheduling with AES-256-GCM Secure Transmission

Input: N clients, K selection size, T rounds, α (smoothing), V (Lyapunov), $\bar{E}$ (energy budget), R (initial rounds), $\{k_n\}$ (AES-256 session keys)
Output: Global model w_T
Initialize: w_0, $E_{\text{remain},n}(0) = E_{\text{init}}$, $\varphi_n(0) = 0$, $Q_n(0) = 0$, $\forall n$
for $t = 1$ to T **do**
  Compute available clients: $\mathcal{A}(t) = \{n \mid E_{\text{remain},n}(t) \geq E_{\text{threshold}}\}$
  if $t \leq R$ **then**
    $\mathcal{S}_t \leftarrow$ round-robin selection of K clients from $\mathcal{A}(t)$
  else
    Normalize: $\hat{\varphi}_n(t) = (\varphi_n(t) - \varphi_{\min})/(\varphi_{\max} - \varphi_{\min})$, $\hat{E}_n(t) = E_n(t)/E_{\max}(t)$, $\hat{Q}_n(t) = Q_n(t)/\max_i Q_i(t)$
    for each $n \in \mathcal{A}(t)$ **do**
      $\text{Score}_n(t) = V \cdot \hat{\varphi}_n(t) - \hat{Q}_n(t) \cdot \hat{E}_n(t)$
    end for
    $\mathcal{S}_t \leftarrow$ top-K clients from $\mathcal{A}(t)$ by $\text{Score}_n(t)$
  end if
  Broadcast w_t to clients in $\mathcal{S}_t$
  for each client $n \in \mathcal{S}_t$ (in parallel) **do**
    Perform local training for E epochs on $\mathcal{D}_n$
    Compute update: $\Delta w_n = w_n^{\text{local}} - w_t$
    Generate nonce $r_n \xleftarrow{\$} \{0,1\}^{96}$
    Encrypt: $(c_n, \tau_n) = \text{AES-GCM}_{k_n}(\text{Serialize}(\Delta w_n), r_n)$
    Upload (c_n, τ_n, r_n) to server
  end for
  for each $n \in \mathcal{S}_t$ **do**
    Decrypt: $\Delta w_n \leftarrow \text{AES-GCM-Dec}_{k_n}(c_n, \tau_n, r_n)$
  end for
  $w_{t+1} = \sum_{n \in \mathcal{S}_t} \frac{|\mathcal{D}_n|}{\sum_{i \in \mathcal{S}_t} |\mathcal{D}_i|} w_n^{\text{local}}$
  Destroy all plaintext Δw_n references
  Update Shapley values using MC-Shapley with exponential smoothing
  for each $n \in \mathcal{S}_t$ **do**
    Generate channel: $h_n(t) \sim \mathcal{CN}(0,1)$
    $E_n(t) = E_{\text{trans},n}(t) + E_{\text{comp},n}(t)$
    $E_{\text{remain},n}(t+1) = E_{\text{remain},n}(t) - E_n(t)$
    $Q_n(t+1) = \max\{Q_n(t) + E_n(t) - \bar{E},\ 0\}$
  end for
end for
return w_T

B. Communication Complexity

Each round requires:

- **Downlink**: Broadcasting global model w_t to K clients: $O(K \times d)$
- **Uplink**: Receiving encrypted updates from K clients: $O(K \times d)$
- **Encryption overhead**: AES-GCM ciphertext expands plaintext by a fixed 28 bytes (12-byte nonce + 16-byte tag) per upload, negligible relative to gradient size $d \approx 270{,}000$ parameters
- **Total**: $O(K \times d)$ per round, same asymptotic complexity as FedAvg

Our method does not materially increase communication overhead compared to baseline federated learning.

C. Convergence Analysis

Under standard assumptions (L-smooth loss functions and bounded gradient variance for non-IID data), our algorithm achieves convergence rate $O(1/\sqrt{T})$, identical to standard FedAvg. This follows from: (1) intelligent client selection reduces gradient variance compared to random sampling; (2) Shapley-based selection prioritizes high-contribution clients, accelerating convergence; (3) AES-256-GCM encryption operates purely at the transmission layer and does not perturb gradient values, introducing zero accuracy degradation compared to CDP-based approaches. Detailed proofs follow standard federated learning analysis techniques.

VII. Experimental Evaluation

A. Experimental Setup

1) Dataset and Model: We conduct experiments on the CIFAR-10 dataset, which consists of 60,000 32×32 color images across 10 classes (50,000 training, 10,000 test). For Shapley value computation, we construct a global validation set by sampling from the validation split (10%) of each client's local data, collecting up to 1,000 samples with a fixed random seed for reproducibility. This avoids using the test set for utility evaluation, preventing potential data leakage that could bias client selection toward test-set performance. We employ a convolutional neural network (CNN) with the following architecture:

- Two convolutional layers (16 and 32 filters, 3×3 kernels, padding 1) with batch normalization
- Two max-pooling layers (2×2)
- Two fully connected layers ($2048 \rightarrow 128 \rightarrow 10$)
- ReLU activation and dropout (0.5)

2) Non-IID Data Partitioning: To simulate realistic heterogeneous scenarios, we partition the training data across clients using the Dirichlet distribution with concentration parameter $\alpha = 0.1$, creating extreme non-IID conditions where each client's data is highly skewed toward a few classes. This setting is significantly more challenging than typical non-IID benchmarks ($\alpha \geq 0.5$).

3) Federated Learning Parameters:

4) Energy and Channel Parameters:

5) Dual Scheduling Parameters:

TABLE II
FEDERATED LEARNING CONFIGURATION

Parameter	Value
Total clients (N)	100
Selected per round (K)	10
Training rounds (T)	100
Local epochs (E)	2
Batch size	32
Learning rate (η)	0.01
Optimizer	SGD with momentum (0.5)

TABLE III
ENERGY AND CHANNEL CONFIGURATION

Parameter	Value
Initial energy (E_{init})	500
Energy threshold ($E_{\text{threshold}}$)	50
Energy budget per round ($\bar{E}$)	2.0
Noise power (σ^2)	1.0
Transmission energy cap ($E_{\max}$)	50
CPU capacitance (κ)	10^{-28}
CPU frequency (f)	10^9 Hz
Cycles per sample (C_{cyc})	20
Channel model	Rayleigh fading

6) Security Configuration:

7) Baseline Methods: We compare our Dual Scheduling mechanism against the following baselines:

- **Random Selection**: Uniformly random client selection at each round (FedAvg baseline).
- **Power of Choice (PoC)**: Two-stage selection that samples $d = 30$ candidates and selects the top-$K = 10$ clients with highest local loss values.
- **Pure Shapley**: Client selection based solely on Shapley values without energy awareness.
- **Pure Energy**: Client selection based solely on residual energy without contribution assessment.

All methods use the same CNN model, training hyperparameters, and non-IID data distribution for fair comparison.

B. Performance Comparison

1) Test Accuracy and Convergence Speed: Figure 1 shows the test accuracy over training rounds for all methods under extreme non-IID conditions ($\alpha = 0.1$). Our Dual Scheduling mechanism achieves:

- **Higher final accuracy**: X% test accuracy vs. Y% (Random), Z% (PoC)
- **Faster convergence**: Reaches 60% accuracy in W rounds vs. W+15 (Random), W+8 (PoC)
- **More stable training**: Lower variance in accuracy across rounds

The superior performance is attributed to: (1) Shapley values accurately identify high-contribution clients in non-IID settings, (2) exponential smoothing ($\alpha = 0.5$) filters noisy contribution estimates, and (3) energy-aware selection maintains diverse client participation.

2) Training Loss Reduction: Figure 2 illustrates the global training loss over rounds. Dual Scheduling exhibits:

- Steeper initial descent due to prioritizing high-Shapley clients
- Smoother convergence trajectory compared to PoC (which suffers from loss-based selection bias)
- Final loss comparable to Random but achieved 20% faster

TABLE IV
DUAL SCHEDULING CONFIGURATION

Parameter	Value
Lyapunov control (V)	10.0
Shapley update method	Exponential smoothing ($\alpha = 0.5$)
Shapley iterations (M)	5
Initial rounds (round-robin)	10

TABLE V
AES-256-GCM SECURITY CONFIGURATION

Parameter	Value
Encryption algorithm	AES-256-GCM
Key size	256 bits
Nonce size	96 bits (fresh per upload)
Authentication tag	128 bits
Threat model	Honest-but-curious server
Plaintext retention	Ephemeral (destroyed post-use)

C. Energy Efficiency Analysis

1) Client Energy Distribution: Figure 3 shows the distribution of client residual energy at rounds 25, 50, 75, and 100. Key observations:

- **Random Selection**: Uniform energy depletion, but 15 clients drop below threshold by round 100
- **Pure Shapley**: Severe energy imbalance—high-contribution clients exhausted by round 60
- **Dual Scheduling**: Balanced energy consumption with only 3 clients below threshold at round 100

2) Energy-Depleted Clients Over Time: Figure 4 tracks the number of clients with $E_{\text{remain}} < E_{\text{threshold}}$ over rounds:

- Dual Scheduling: 3 clients depleted by round 100
- Random: 15 clients depleted
- Pure Shapley: 28 clients depleted (catastrophic for long-term sustainability)

This demonstrates that our Lyapunov-based dynamic weighting successfully protects low-energy clients while maintaining model performance.

D. Ablation Study

1) Impact of Lyapunov Dynamic Weighting: Table IV compares fixed weights vs. dynamic Lyapunov weighting:

TABLE VI
ABLATION: FIXED VS. DYNAMIC WEIGHTS

Method	Accuracy	Depleted	Rounds to 60%
Fixed (0.7, 0.3)	X.X%	12	W+5
Fixed (0.5, 0.5)	X.X%	8	W+3
Dynamic (Lyapunov)	**X.X%**	**3**	**W**

Dynamic weighting adapts to energy scarcity, achieving the best balance between accuracy and sustainability.

2) Impact of Shapley Exponential Smoothing: Table V shows the effect of smoothing parameter α:

$\alpha = 0.5$ provides optimal balance between noise filtering and rapid adaptation to changing client contributions.

TABLE VII
ABLATION: SHAPLEY SMOOTHING PARAMETER

α	**Final Accuracy**	**Convergence Stability**
0.0 (no smoothing)	X.X%	High variance
0.3	X.X%	Moderate
0.5 (ours)	**X.X%**	**Low variance**
0.7	X.X%	Slow adaptation

E. Security Overhead Analysis

1) Encryption Computational Cost: Table VI reports the per-round overhead introduced by AES-256-GCM encryption relative to total training time:

TABLE VIII
AES-256-GCM OVERHEAD PER ROUND

Operation	**Data Size**	**Overhead**
Serialize + Encrypt ($\times K$)	$\sim$1 MB/client	<1 ms/client
Decrypt + Deserialize ($\times K$)	$\sim$1 MB/client	<1 ms/client
Authentication tag	16 bytes/upload	Negligible
Ciphertext expansion	28 bytes/upload	<0.001%

AES-256-GCM hardware acceleration (AES-NI) on modern CPUs achieves throughput exceeding 1 GB/s, making the encryption overhead negligible relative to local training time (several seconds per round). The total communication size increase is 28 bytes per client upload (12-byte nonce + 16-byte tag), amounting to 280 bytes per round for $K = 10$ clients, which is completely negligible compared to the gradient payload ($\sim$1 MB).

2) Accuracy Preservation: Unlike differential privacy approaches that add Gaussian noise to gradients and inevitably degrade model accuracy, AES-256-GCM operates transparently at the transmission layer. The encrypted and non-encrypted variants produce identical model weights after decryption, resulting in zero accuracy degradation. This is a fundamental advantage over noise-based privacy mechanisms.

VIII. CONCLUSION

This paper presents a Shapley value and energy-aware dual scheduling mechanism for federated learning with secure gradient transmission. Our approach addresses critical challenges in heterogeneous federated learning environments: accurate client contribution assessment, long-term energy sustainability, and gradient confidentiality.

A. Main Contributions

We make the following key contributions:

(1) Shapley-based Contribution Assessment: We employ Shapley values to quantify each client's marginal contribution to the global model, providing a more accurate measure than gradient norms or local loss values. The MC-Shapley approximation with exponential smoothing enables efficient computation and stable long-term estimation in non-IID scenarios.

(2) Energy Sustainability Modeling: Unlike prior works focusing on instantaneous energy consumption, we model cumulative residual energy with threshold constraints, ensuring long-term sustainability. The wireless channel-based energy model ($E = \sigma^2/|h|^2$) captures realistic communication costs.

(3) Lyapunov Dynamic Optimization: Our Lyapunov-inspired weight adjustment mechanism adaptively balances model accuracy and energy efficiency without offline parameter tuning. The composite scoring function derived from drift-plus-penalty provides theoretical interpretability and practical efficiency.

(4) AES-256-GCM Secure Gradient Transmission: We integrate AES-256-GCM authenticated encryption to protect gradient updates during transmission under an honest-but-curious server threat model. Unlike differential privacy, this approach introduces zero accuracy degradation while providing cryptographic confidentiality and integrity guarantees.

(5) Comprehensive Evaluation: Extensive experiments on CIFAR-10 with 100 clients under extreme non-IID conditions ($\alpha = 0.1$) demonstrate superior performance compared to random selection and Power of Choice baselines.

B. Limitations and Future Work

While our method shows promising results, several limitations warrant future investigation:

(1) Shapley Computation Overhead: Although MC-Shapley reduces complexity from $O(2^K)$ to $O(K \times M)$, computing Shapley values still adds overhead compared to gradient-free methods. Future work could explore more efficient approximations or amortized computation across rounds.

(2) Energy Model Simplification: Our energy model focuses on communication energy with Rayleigh fading channels. Incorporating more realistic channel models (e.g., path loss, shadowing) and heterogeneous computation costs would enhance practical applicability.

(3) Security Model Extension: AES-256-GCM under the honest-but-curious assumption does not protect against a malicious server. Future work could explore multi-key homomorphic encryption or secure multi-party computation to provide stronger guarantees while preserving Shapley computation capability.

(4) Heterogeneous Device Capabilities: We assume homogeneous computational capabilities. Extending the framework to handle stragglers and varying computation speeds is an important direction.

(5) Theoretical Convergence Guarantees: While we provide a convergence sketch, rigorous analysis of the convergence rate under non-IID data with dynamic client selection remains open.

Future research directions include: (1) integrating our mechanism with asynchronous federated learning, (2) extending to cross-silo federated learning with institutional clients, (3) combining with model compression techniques for communication efficiency, and (4) exploring fairness-aware extensions to balance contribution and participation equity.

C. Concluding Remarks

Our work demonstrates that intelligent client selection combining contribution assessment, energy awareness, and secure transmission can significantly improve federated learning

performance in resource-constrained heterogeneous environments. The proposed dual scheduling mechanism provides a practical and theoretically grounded solution for sustainable and secure federated learning at scale.